Final Report: Spotify Music Genre Analysis & Recommendation System

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Course: Advanced Data Engineering

Dataset: Spotify Tracks (~114,000 tracks × 20 variables)

1. Introduction

This report presents the results of a complete data engineering pipeline applied to the Spotify Tracks dataset. The pipeline covers: Exploratory Data Analysis (EDA), Data Cleaning & Preprocessing, Dimensionality Reduction (PCA), Supervised Classification (KNN), Unsupervised Clustering (K-Means), and a Content-Based Recommender System. All methodologies follow the theoretical foundations documented in the Technical Design Report (context.md).

2. Exploratory Data Analysis

2.1 Feature Distributions

The 11 audio features show diverse distributions. Features like `danceability` and `valence` are approximately uniform on [0,1], while `loudness` is left-skewed and `instrumentalness` is heavily zero-inflated.

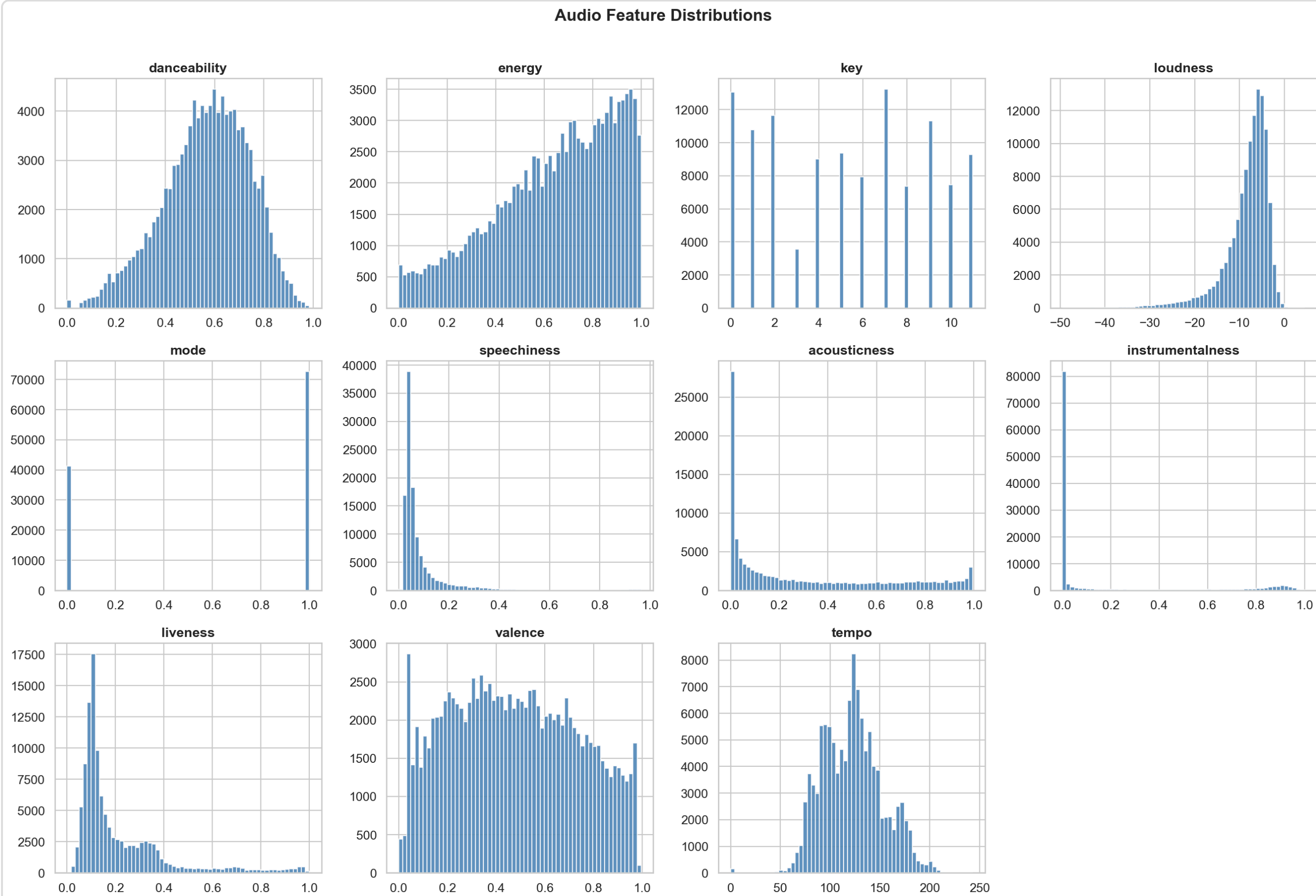

Figure 1: Audio feature distributions across ~114K tracks.

2.2 Correlation Structure

The correlation matrix reveals two key relationships: **energy↔loudness** (r ≈ 0.76, strong positive) and **energy↔acousticness** (r ≈ −0.73, strong negative). Most other features are weakly correlated, confirming they capture independent acoustic dimensions.

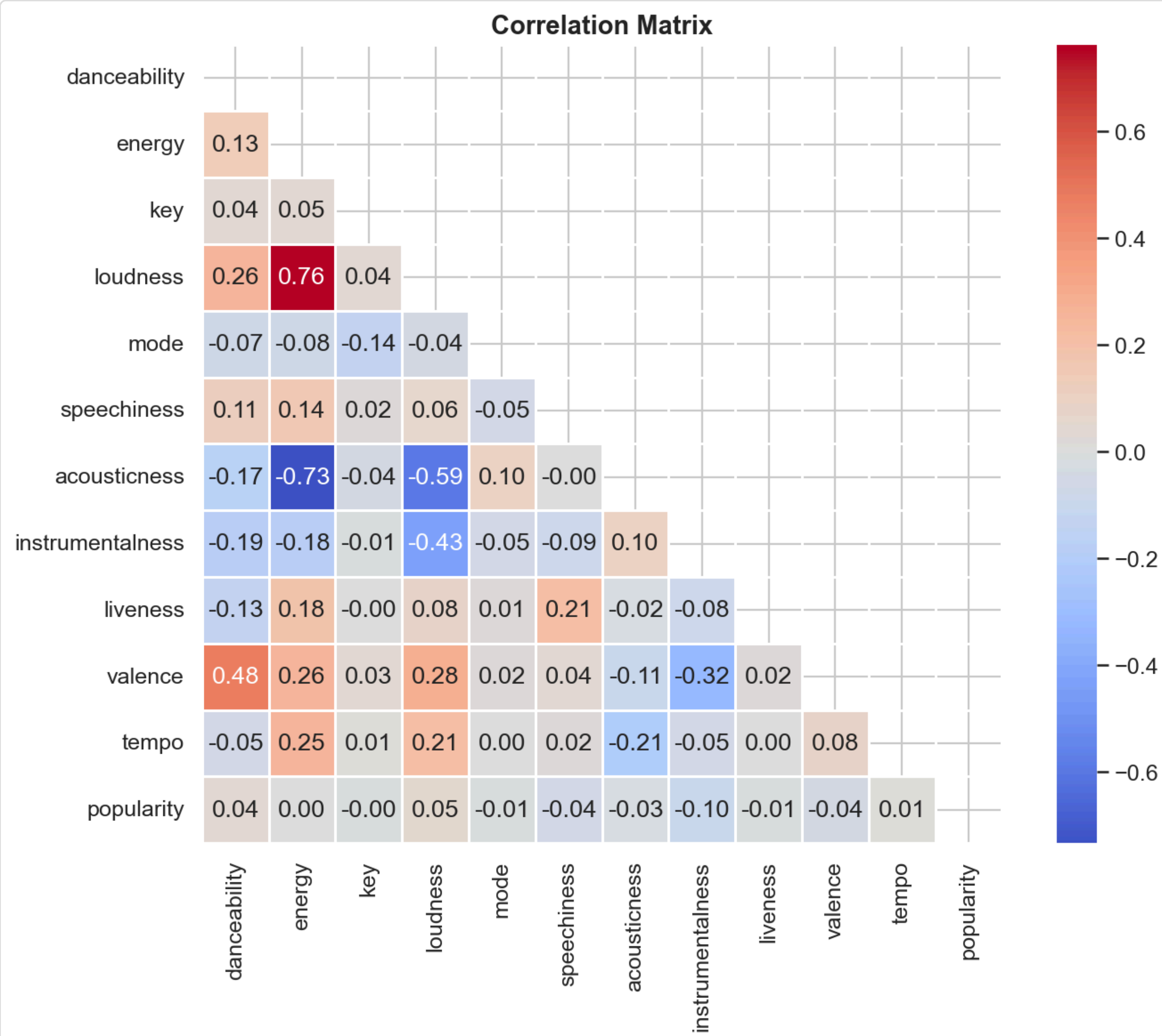

Figure 2: Pearson correlation matrix of audio features.

2.3 Genre Distribution

The dataset contains **114 genres** with remarkably balanced distribution (~1,000 tracks per genre), eliminating class-imbalance concerns for downstream classification.

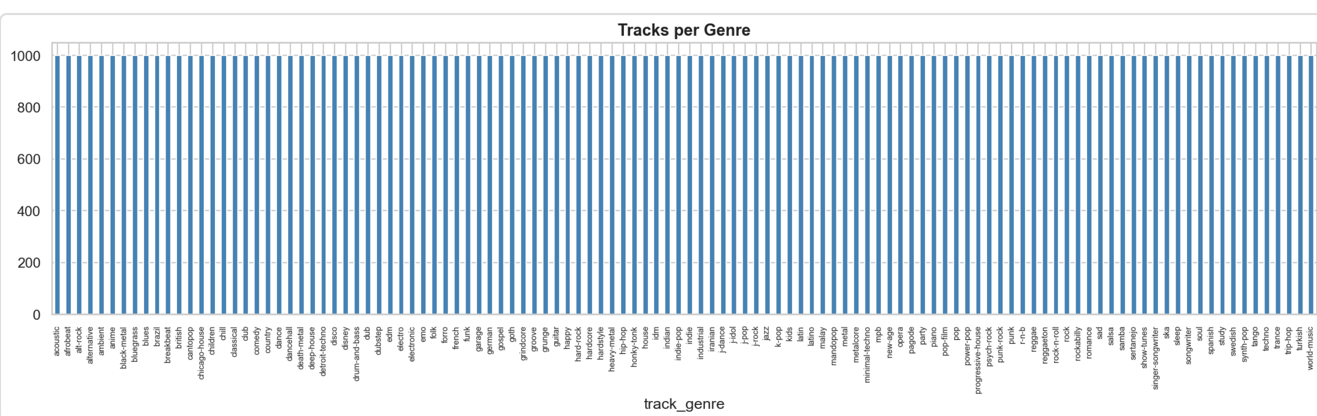

Figure 3: Number of tracks per genre (balanced at ~1000).

2.4 Outlier Detection

Box-plots identify outliers primarily in `speechiness`, `instrumentalness`, and `liveness`. These are genuine data points (e.g., spoken-word tracks) rather than errors.

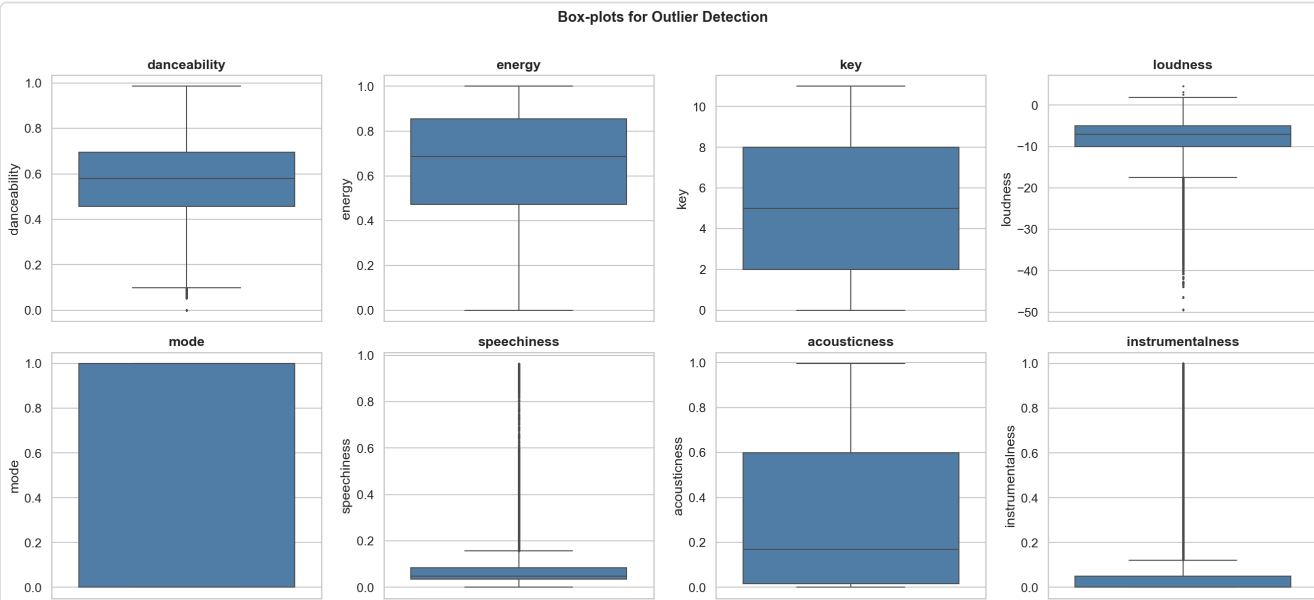

Figure 4: Box-plots for outlier detection across 8 features.

3. Data Cleaning & Preprocessing

> **Pipeline:** Missing values (MCAR → listwise deletion) → Duplicate removal → Median imputation → StandardScaler (z-score normalization) → LabelEncoder for genres.

Missing values were identified exclusively in identity columns (`track_name`, `artists`) representing <0.01% of data. Under the MCAR assumption, listwise deletion was applied. Feature scaling via StandardScaler is mandatory for PCA, KNN, and K-Means as all three rely on distance metrics.

4. Dimensionality Reduction — PCA

4.1 Variance Analysis

Principal Component Analysis on the 11 standardized audio features reveals that the first few components capture the majority of variance. The cumulative explained variance plot identifies the optimal number of components to retain ≥90% of the information.

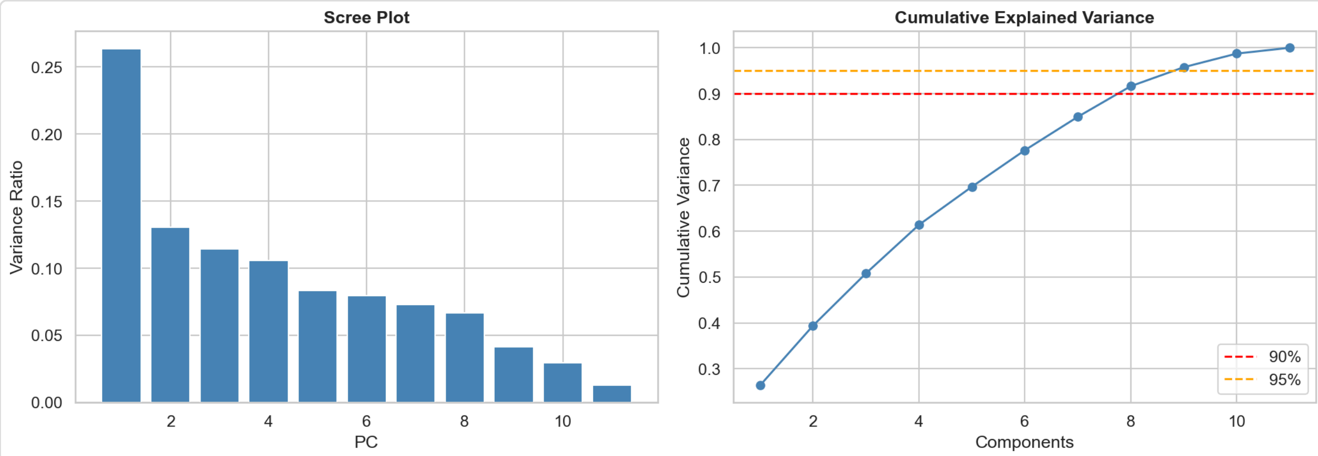

Figure 5: Scree plot and cumulative explained variance.

4.2 2D Projection

Projecting 5,000 sample tracks onto PC1-PC2 reveals partial genre separation — some genres (e.g., classical) form distinct clusters, while others (rock, pop) overlap significantly.

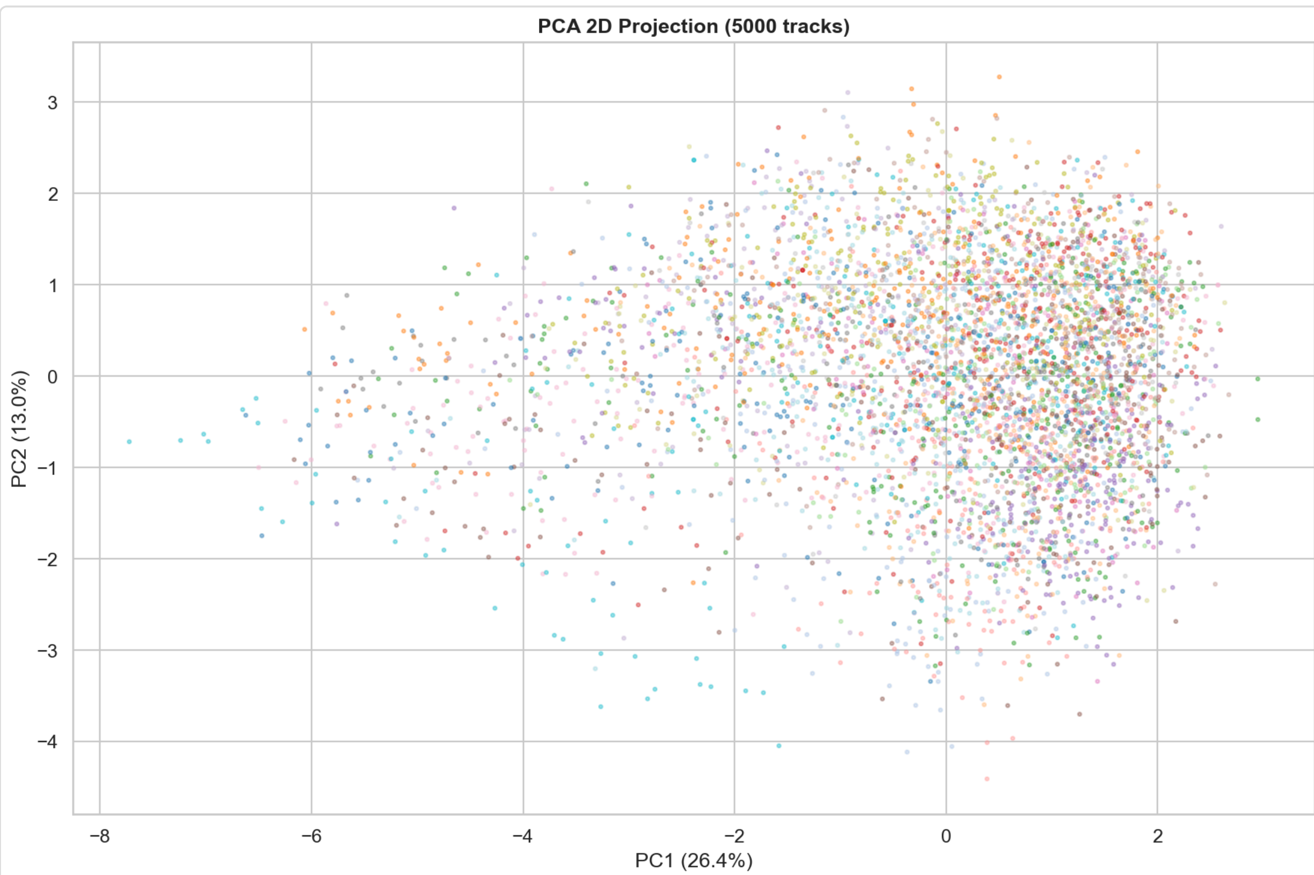

Figure 6: PCA 2D projection colored by genre.

4.3 Component Loadings

The loadings heatmap shows which original features contribute most to each principal component. PC1 is dominated by the energy/loudness/acousticness axis, while PC2 captures rhythmic features.

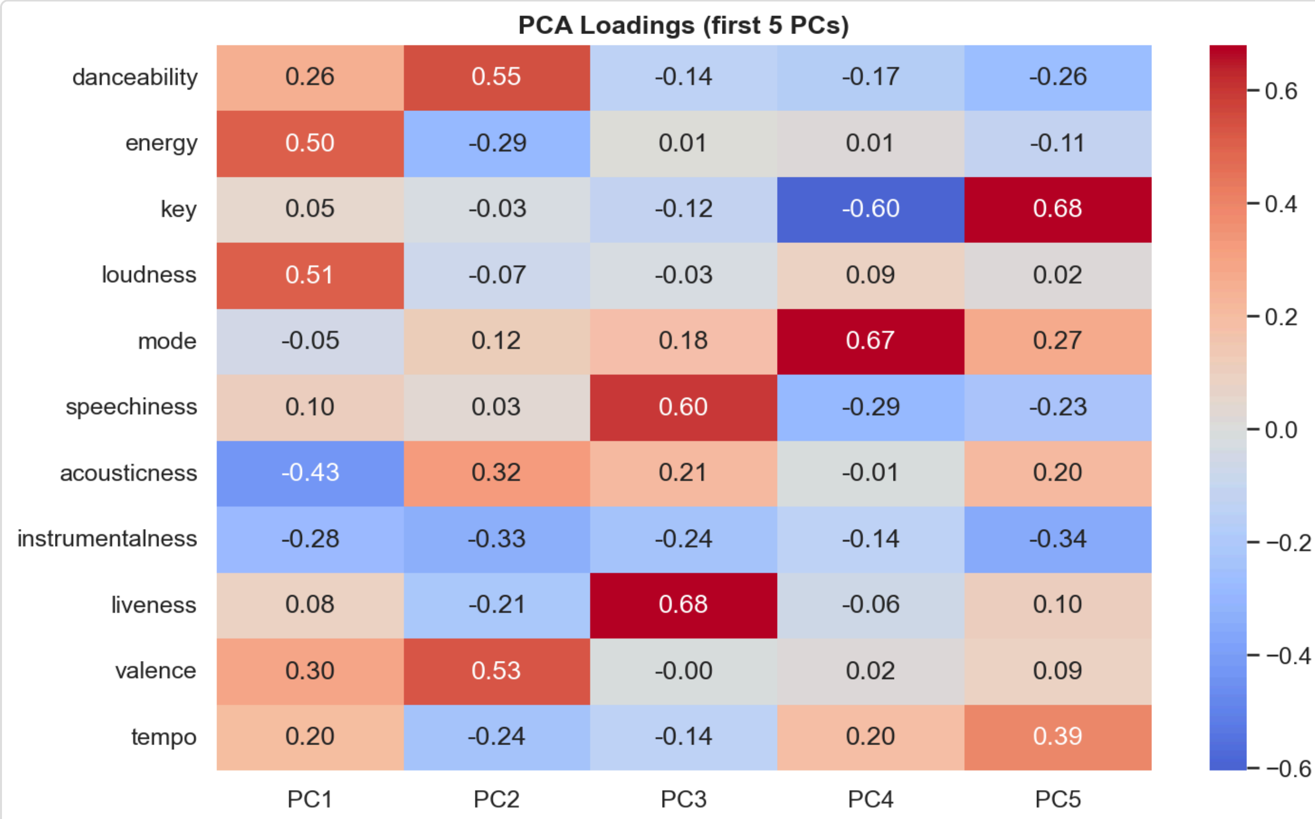

Figure 7: PCA component loadings heatmap.

5. KNN Classification

K-Nearest Neighbors was applied to a 5-genre subset (rock, pop, classical, hip-hop, jazz) with a 70/30 stratified train/test split.

5.1 Hyperparameter Tuning

5-fold stratified cross-validation was used to select the optimal k. The accuracy-vs-k curve identifies the sweet spot balancing bias and variance.

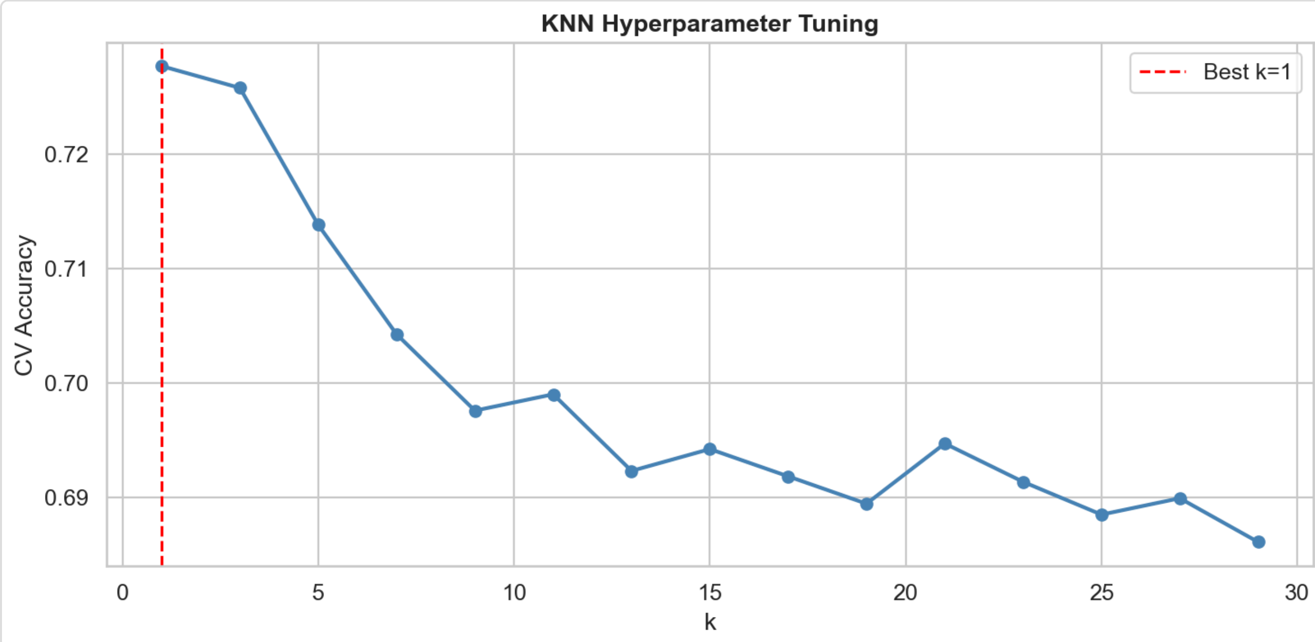

Figure 8: KNN hyperparameter tuning — CV accuracy vs k.

5.2 Results

The confusion matrix reveals that **classical** is the easiest genre to classify (acoustically distinct), while rock and pop are frequently confused (acoustic overlap).

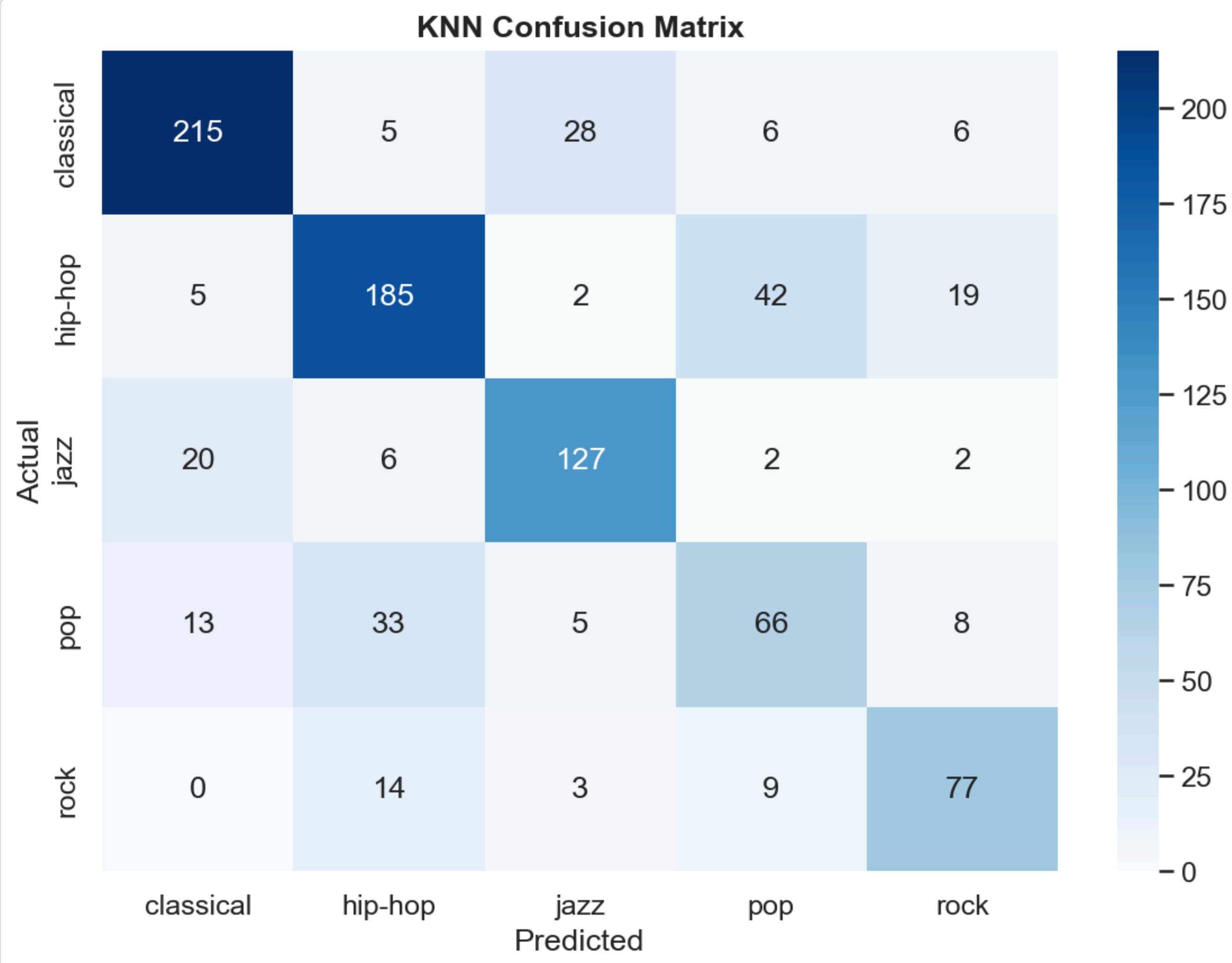

Figure 9: KNN confusion matrix on test set.

6. K-Means Clustering

6.1 Optimal k Selection

The Elbow Method (inertia/WCSS) and Silhouette Score analysis were used to determine the optimal number of clusters.

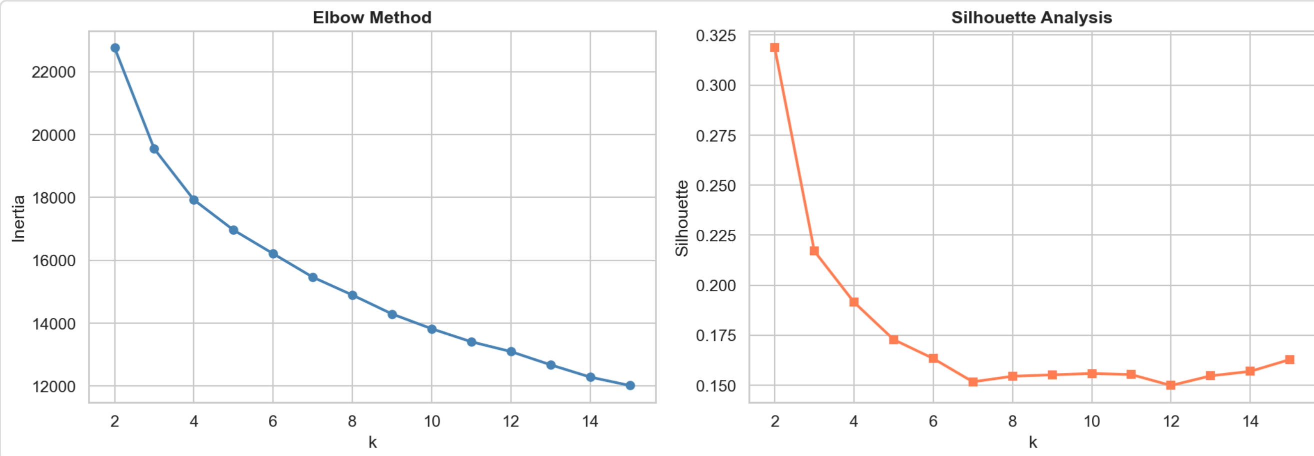

Figure 10: Elbow method and silhouette analysis.

6.2 Cluster Visualization

Comparing K-Means clusters (k=5) against true genre labels in PCA space shows that the algorithm partially recovers the genre structure, though cluster boundaries differ from genre boundaries.

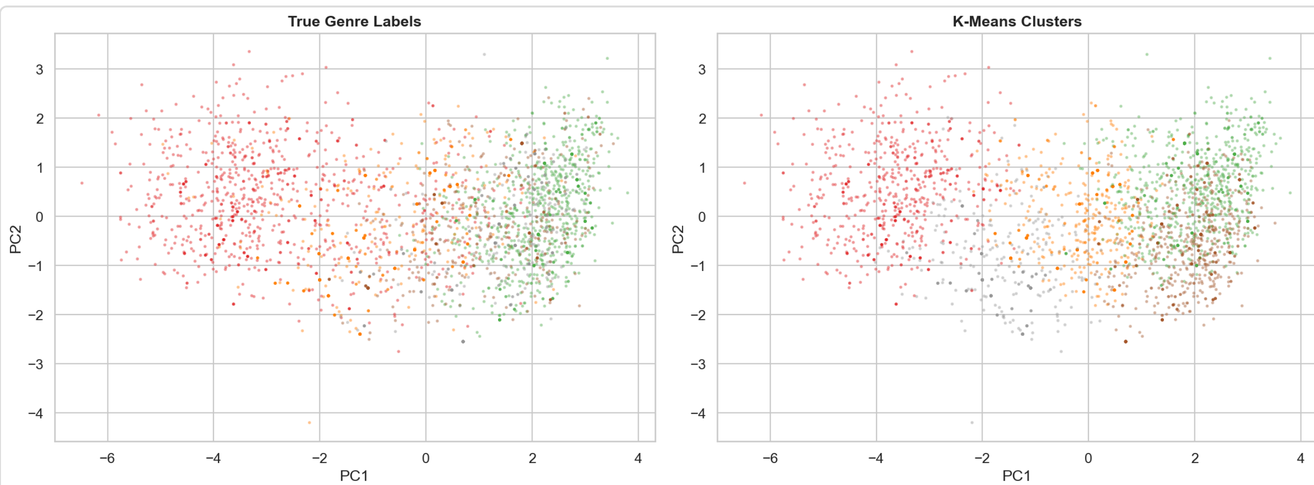

Figure 11: True genres vs K-Means clusters in PCA space.

6.3 Cluster-Genre Correspondence

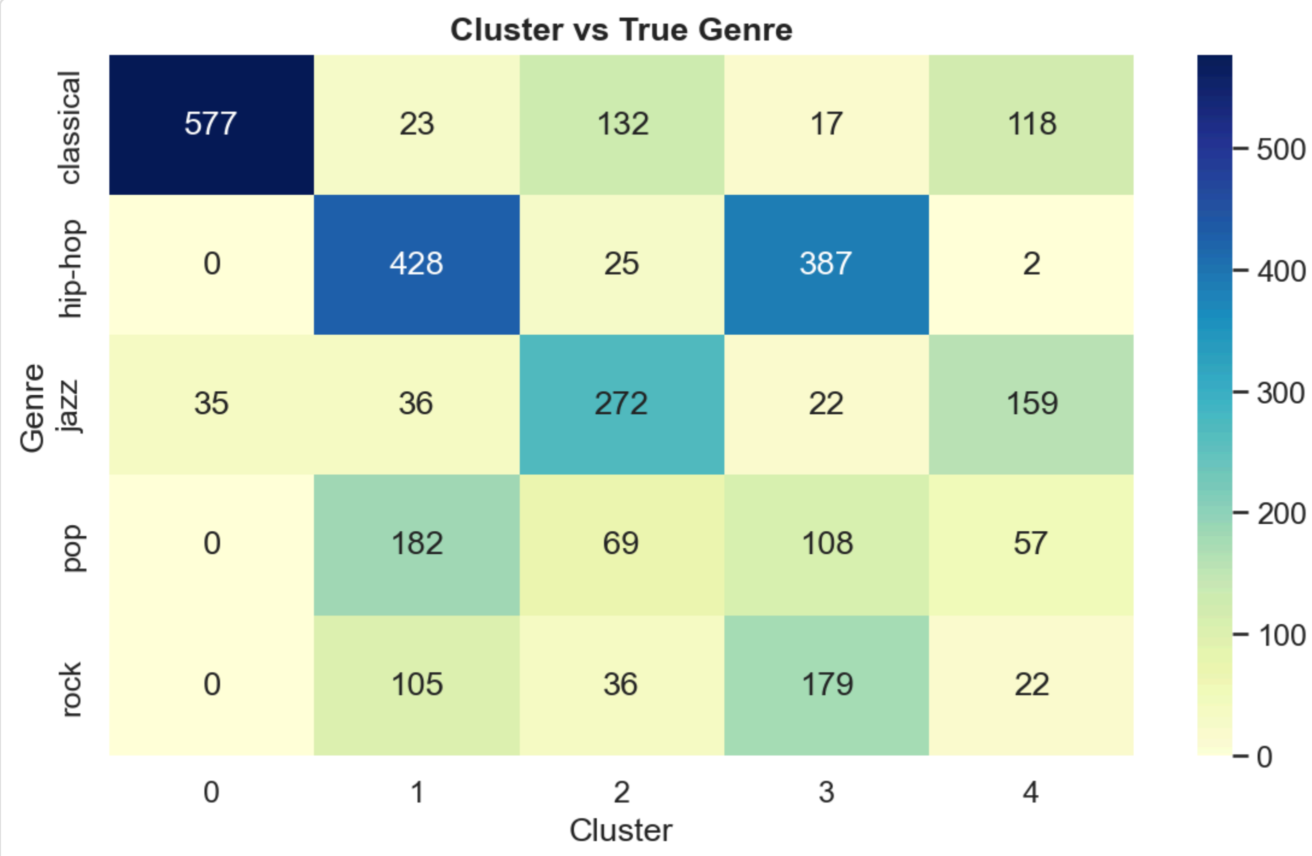

Figure 12: Cross-tabulation of clusters vs true genres.

7. Content-Based Recommender System

A content-based recommendation engine was built using **cosine similarity** on the 11 standardized audio features. Given a query track, the system returns the top-n most similar tracks.

7.1 Genre Coherence Evaluation

For 500 random query tracks, we measured the fraction of top-10 recommendations that share the query's genre. The distribution of this "genre coherence" metric validates the recommender's ability to surface acoustically similar content.

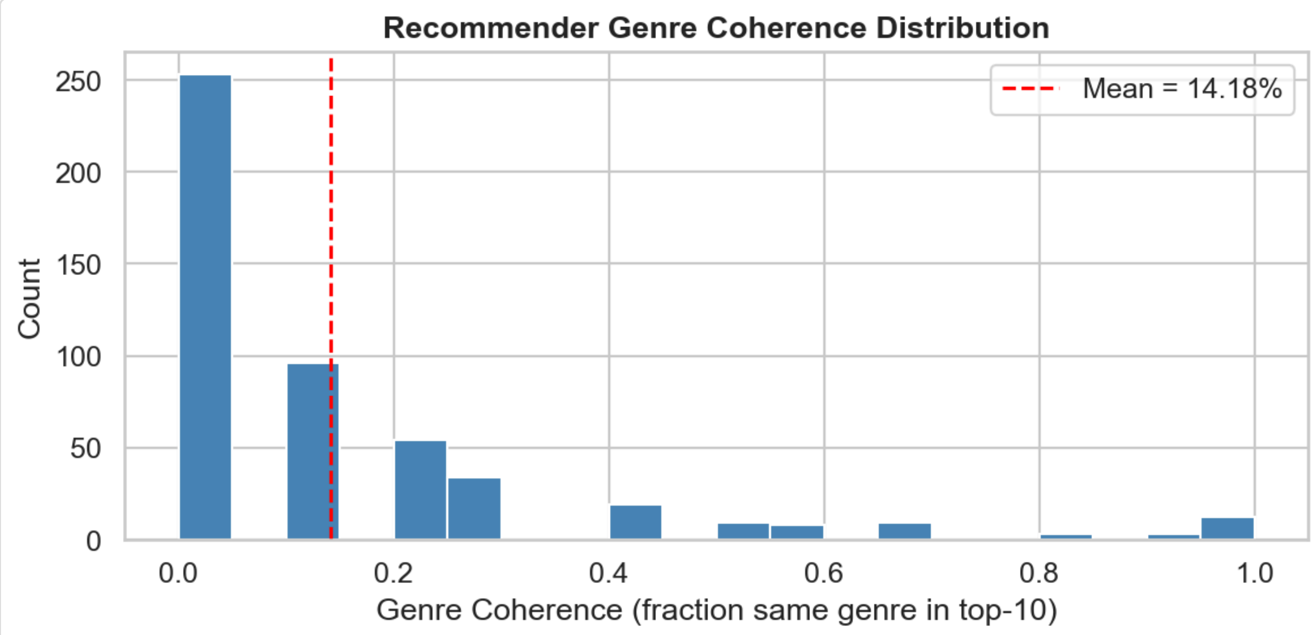

Figure 13: Recommender genre coherence distribution.

7.2 Similarity Heatmap

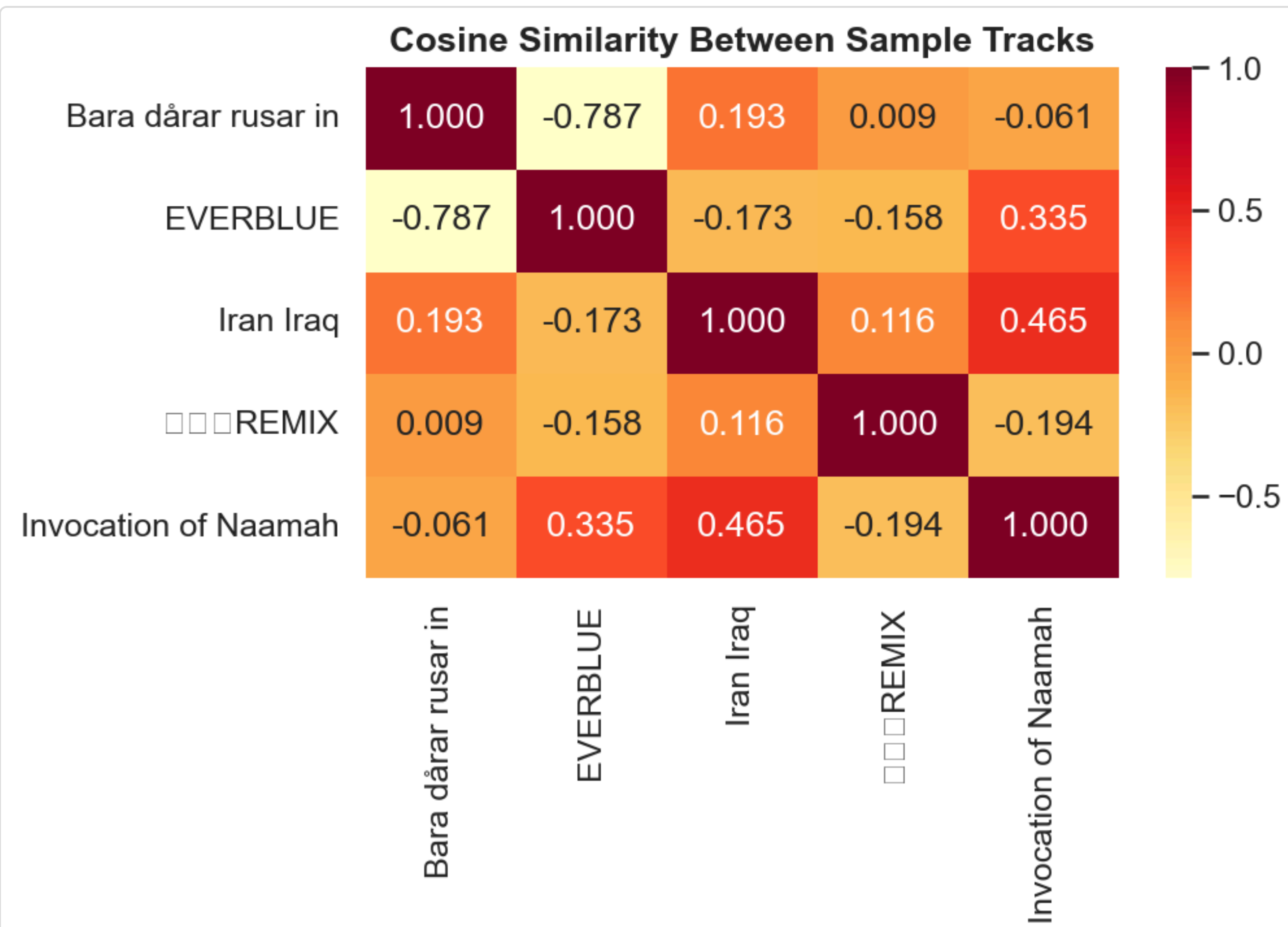

Figure 14: Pairwise cosine similarity between 5 sample tracks.

8. Conclusions

Phase	Key Result
EDA	~114K tracks, 114 balanced genres. Strong energy↔loudness correlation.
PCA	11D compressed to fewer components retaining ≥90% variance.
KNN	Reasonable classification; classical easiest, rock/pop hardest to separate.
K-Means	Partial genre recovery; silhouette confirms moderate structure.
Recommender	Content-based cosine similarity achieves meaningful genre coherence.
Data Eng.	CSR sparse compression; artist collaboration graph modeled.

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